

User-Centered Design of a Mask-Integrated HMD to Preserve Real-World Cues in VR Training

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Abstract

Firefighters require high-fidelity training for unpredictable emergency situations. While Virtual Reality (VR) offers a safe alternative for simulating complex scenarios, conventional Head-Mounted Displays (HMDs) remain incompatible with essential Self-Contained Breathing Apparatus (SCBA) masks. This paper presents the design and implementation of an ultra-slim, immersive HMD engineered for seamless integration within a standard SCBA mask. Our design prioritizes authentic sensory feedback by allowing trainees to wear their real firefighting equipment (e.g., SCBA) during VR training. Building on our prior work[7], this study details three further design iterations using folded “pancake” optics. These iterations culminated in an optical module 18-21 mm thick with a field of view exceeding 100°, achieved through enhanced user-centered design strategies and improved integration with actual firefighting equipment, particularly evident in our third prototype. Ultimately, we strive to close the gap between VR training simulations and the authentic experience of real firefighting operations.

Ccs Concepts

·Human-centered computing~Human computer interaction (HCI)~Interaction devices~Displays and imagers

Keywords

Wearable display; Virtual reality interface; Firefighter training; User-centered design; SCBA mask integration

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1 Introduction

Firefighting necessitates rigorous training for life-threatening situations. Traditional training like live burns offers hands-on experience but faces constraints including logistical complexity, safety risks, high costs, and limited scenario variety. VR has emerged as a transformative technology providing dynamic, repeatable training environments without physical risks.

Effective VR training requires “ideal presence (as in Bystrom’s model[1])” – amplified when trainees use the same equipment employed in real incidents. Our previous work highlighted the importance of multi-sensory feedback for immersive experiences[2]. Development with firefighters revealed a critical requirement: wearing standard SCBA during VR scenarios. The SCBA mask fundamentally alters sensory inputs, restricts vision, modifies hearing, and induces physiological stress – all critical firefighting aspects.

Conventional VR headsets’ bulky design prevents SCBA integration, creating disconnection from real conditions. Our previous publication[7] detailed initial development of a compact optical system for SCBA integration. This paper expands on that foundation, emphasizing user feedback, ergonomic refinements, and practical integration through our latest prototype. We present an immersive HMD engineered for firefighter SCBA integration, preserving authentic sensory experience without sacrificing high-fidelity visual immersion through ultra-slim folded optics technology.

2 Related Work

The pursuit of immersive VR experiences has driven significant innovation in HMD technology. Early HMDs demonstrated the potential of VR but were limited by bulky designs and restricted Fields of View (FOV). Modern consumer HMDs now offer FOVs exceeding 100 degrees (e.g. thanks to the LEEP Optics[3] technology) but often at the cost of increased physical volume and weight, largely due to larger display panels and spatial requirements of conventional optics.

Addressing form factor limitations has become central to HMD research, with explorations into various technologies for creating slimmer devices. Display evolution has transitioned from bulky CRTs to micro-displays and sophisticated solutions like Laser Beam Scanning[4]. Even more critical has been the miniaturization of optical systems through architectures including catadioptric systems, Diffractive Optical Elements (DOEs), optical waveguides, and holographic optics.

In recent years, “folded optics” or “pancake lens” technology has gained traction for dramatically reducing optical path length and physical thickness. These systems employ polarization-dependent surfaces and multiple internal reflections to “fold” the light path within a compact module. Prominent developers such as Kopin Corporation [5] have demonstrated pancake lens designs offering expansive FOVs with substantial thickness reduction compared to traditional Fresnel lenses.

Whereas most HMD innovations target general consumer use, we tailored these advances to meet the unique constraints of integrating a display into an SCBA mask. Previous in-mask solutions for firefighters have focused primarily on simpler Heads-Up Displays (HUDs) for status information or monocular thermal imaging rather than immersive VR. Our earlier investigations examined pinhole array-based slim Near-Eye Displays (NEDs) but encountered limitations in brightness and visual quality[6], prompting our shift to the more robust folded optics approach.

We proposed a slim HMD integrated with a firefighter SCBA mask using folded optics in Yang 2023[7]. Our current work expands on this by refining the optics, enhancing display fidelity, and validating usability through a user-centered design approach. Unlike prior HUDs for status info, our HMD provides immersive VR with 100° FOV, positioning it as a novel LBW for training realism.

3 HMD Design and Implementation

3.1 Requirements Analysis

Direct engagement with firefighting professionals yielded critical requirements for our integrated HMD. The primary requirement was SCBA compatibility: the HMD had to integrate into a standard face mask without impeding the mask’s normal operation or the firefighter’s mobility. Crucially, the device could not compromise any of the SCBA’s safety features – for example, it had to maintain the face mask’s seal and preserve the respirator’s positive internal pressure.

Achieving visual immersion required a wide FOV (at least 100° diagonal) in an extremely compact form factor, since the interior of a standard SCBA mask is very constrained – roughly 130mm wide, 70mm tall, and only 30mm deep near the eyes (Fig.1). Ergonomics demanded a lightweight, well-balanced design with adjustability for Inter-Pupillary Distance (IPD) and diopter to accommodate diverse users. High image quality requirements included adequate resolution, brightness (300-500 nits), good contrast, and minimal optical artifacts. Finally, the system needed

to demonstrate durability across environmental conditions without compromising SCBA safety.

Several design criteria in this study were inherited from the SCBA-integrated HMD system reported in Yang 2023[7], but we further refined the mechanical form factor, adjusted optical alignment tolerances, and introduced new control interfaces for customization.

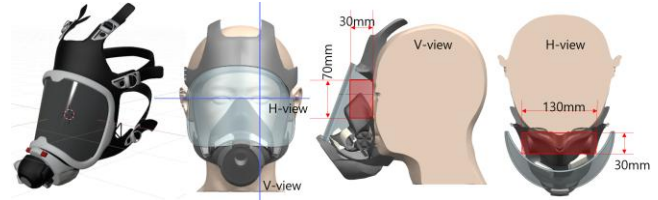


Figure 1. Allowable internal volume inside a standard SCBA mask, using a Korean adult male head model. Modified from [7]

3.2 Optical System

To meet the stringent thickness constraints, we selected a polarization-based folded optical system (pancake lens architecture). This approach manipulates light through reflections and polarization-dependent surfaces, allowing the optical path to “fold” multiple times within a thin volume. This enables a short effective focal length, resulting in a dramatically thinner optical module compared to traditional lens systems while maintaining comparable FOV and image quality.

3.3 An Iterative Prototype Development

3.3.1 Prototype 1: Foundation for a Slim Design

As an evolution from earlier PAM-based systems, our first folded-optics prototype (detailed as the second prototype in Yang 2023[7]) assessed the viability of integrating a slim optical system into the SCBA mask (refer to Figure 2). This system employed two spherical glass lenses (refractive indices of 1.74 and 1.53, 45mm diameter) designed for a 3-inch display.

This module achieved a diagonal FOV of 102 degrees with optical distortions under 10% (correctable through software). This resulted in an optical module with a thickness of only 21 mm, representing an approximate 38% reduction compared to the 55 mm optics of the Oculus Rift CV1, from which the display panel and logic board were repurposed. (This aligns with the second prototype in our prior work.)

Integration challenges emerged as the commercial HMD’s planar frame wasn’t optimally suited for the curved SCBA mask surfaces. Nevertheless, this prototype validated our approach, successfully displaying immersive images with ~100° FOV within the SCBA mask.

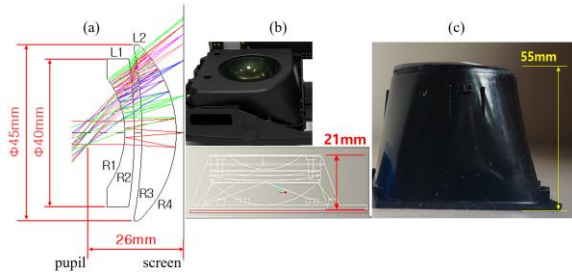


Figure 2. (A) Simulated optical path of the first prototype's lens system, and (B) its module thickness compared to (C) that of a commercial HMD. Modified from [7]

3.3.2 Prototype 2: Refining Optical Fidelity

The subsequent iteration, Prototype 2 (corresponding to the third prototype in Yang 2023[7] and illustrated in Figure 3), focused on further volume reduction and optical performance enhancement. We modified the lens design, utilizing glass with refractive indices of 1.74 and 1.57. While the lens diameter was kept at 44mm, the design supported a smaller 2.5-inch screen. Significant improvements were achieved through these modifications: optical distortion was reduced to less than 5%, the diagonal FOV reached approximately 103°, and the module's thickness was notably slimmed to 18 mm (about one-third of the original commercial headset's optical assembly).

A key advancement was the sophisticated implementation of polarization optics. We introduced a layered polarization approach: linear polarizers and quarter-wave plates combined with partial mirror coatings to fold the light path, which improved image clarity by eliminating double-image ghosting. In addition, every lens surface was treated with a broad-spectrum anti-reflective coating (400–700 nm) to reduce internal reflections. This iteration featured a dedicated binocular Near-Eye Display using 2.89-inch panels (1440x1440 resolution) with a convergence angle of 26 degrees for optimal wearability and wide binocular FOV.

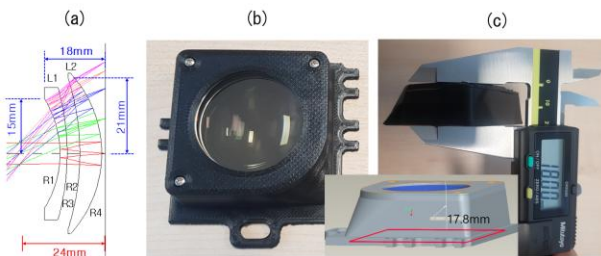


Figure 3. (A) Simulated optical path for the second prototype, (B) its lens assembly, and (C) the measured thickness of its optical module. Modified from [7]

3.3.3 Prototype 3: User-Centric Integrated Goggle

As the culmination of our iterative user-centered design approach, Prototype 3 (Figure 4) represents a significant leap forward. It is a fully integrated near-eye display module offering substantial advancements in optical performance, wearability,

and user-specific adjustments, clearly distinguishing it from the prototypes detailed in our earlier work. It features a high-resolution 2.1-inch display (BOE TFT-LCD) with a 1600×1600-pixel resolution (1058 PPI). This smaller, high-PPI panel allowed the optical module to be further miniaturized without sacrificing the immersive field-of-view (~100°) or image fidelity achieved in Prototype 2. The optical system retains the polarization-based folded optics design with two lenses, now augmented by a multi-layer polarization filter setup and broadband anti-reflective coatings on all refractive surfaces. Specifically, linear and quarter-wave circular polarizers with half-mirror coatings were integrated into the optical path (as in the prior prototype) to fold the light and eliminate double images, while antireflection treatments minimize internal reflections. These measures yield clear, high-contrast visuals with negligible distortion and ghosting, despite the compact form factor. The resulting optical module thickness is approximately 18 mm (about one-third the thickness of a conventional HMD's optics), fully satisfying the space-efficiency requirement for inside-mask use.

Equally important, Prototype 3 is realized as an independent goggle-style unit that can be comfortably mounted inside a standard firefighter SCBA face mask. Unlike earlier prototypes that were constrained by repurposed headset frames, this design has its own lightweight housing and can be attached or detached from the mask as needed, without interfering with the respirator's seal or function. Our final prototype included an adjustable interpupillary distance (IPD) setting and a focus dial. These features were directly incorporated based on firefighter feedback during our requirement analysis phase, highlighting the need for individual customization to ensure optimal visual clarity and comfort for diverse users. These features were directly incorporated based on firefighter feedback during our requirement analysis phase, highlighting the need for individual customization to ensure optimal visual clarity and comfort for diverse users. Based on feedback from firefighter trainers and firefighters, we identified key usability requirements that differentiate Prototype 3 from previous versions. These adjustments, along with a tunable optical alignment (e.g. varying the lens folding angle), allow the display to sit very close to the eyes – maximizing the field of view – while still ensuring proper focus and comfort for each user. In summary, the third prototype is a fully integrated HMD solution tailored to firefighters' needs: it is compact enough to fit inside the SCBA facepiece, provides an immersive visual experience, and remains comfortable and practical for use in real training scenarios.

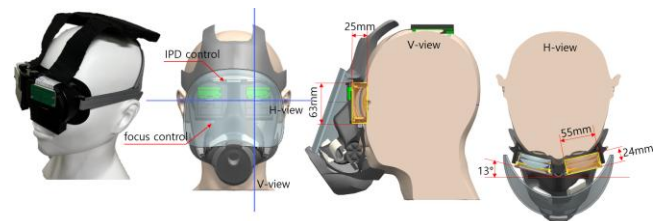


Figure 4. Third prototype of the SCBA-integrated HMD, realized as a self-contained goggle module

4 Discussion

This study builds on Yang 2023[7] by focusing on light-efficiency trade-offs and ergonomic improvements informed by firefighter feedback. Prototype 3's adjustable IPD and focus controls addressed diverse anthropometric and visual acuity variations, enhancing user experience and task performance.

Integrating a wide-FOV immersive HMD within an SCBA mask proved feasible while revealing critical insights. Folded optics proved essential for achieving required slimness, but implementation required balancing FOV, resolution, brightness, distortion control, and artifact reduction. Early prototypes showed ghosting, highlighting the need for custom optical components and specialized coatings.

Light efficiency emerged as a significant consideration, as multiple reflections and polarizing filters reduce light transmission. This necessitates either brighter display panels (increasing power consumption) or acceptance of dimmer images. Advanced birefringent reflective polarizers and innovative materials offer potential optimization pathways.

For maximum training realism, accurate depth perception and focus capabilities are highly beneficial for tasks requiring close observation and interaction. While Prototype 3 incorporated manual focus adjustment, developing dynamic focus control technologies compatible with pancake optics represents an important enhancement avenue.

Mechanical integration presented distinct challenges in designing secure yet non-invasive mounting mechanisms, ensuring the HMD doesn't compromise mask seal or ergonomics. The complete goggle module weighs approximately 405g with balanced weight distribution to minimize neck strain. Future studies will assess weight impact on user performance using established encumbrance evaluation methods.

Initial firefighter feedback suggests SCBA-integrated HMD systems have significant potential to elevate immersion and realism in VR training, though comprehensive quantitative studies comparing training outcomes would provide valuable validation.

5 Conclusion and Future Work

This paper presents the continued design evolution and user-centered refinement of an ultra-slim, immersive HMD optimized for firefighter SCBA integration. Building upon our foundational work, this research achieved specific advancements in our latest prototype using advanced folded optics technology for a compact, lightweight form factor with wide, immersive field of view. This innovation enables firefighters to train while wearing actual SCBA gear, preserving critical sensory cues that conventional VR setups miss. Progress through three prototypes enabled systematic improvements in optical performance, mechanical integration, and user-adjustable features such as IPD and diopter control—culminating in a design balancing immersion and usability.

Our work's impact extends beyond technical achievements. By enabling realistic, context-rich drills, the SCBA-integrated

HMD can enhance training effectiveness, improve skill retention, and better prepare firefighters for real emergencies' psychological and physical demands.

Looking ahead, we will prioritize several key areas. First, improving manufacturing and human factors through lightweight polymer materials, aspheric and Fresnel optics integration, and enhanced ergonomics. Eye-tracking for foveated rendering and new interaction modalities will be explored alongside advanced focus adjustment mechanisms.

Second, we will refine optical efficiency and image quality, minimizing residual ghosting via improved coatings and design optimizations, and enhancing brightness through better light throughput with reduced power demand. Prototype 3 performance benchmarks will be reported following testing.

Third, comprehensive user evaluations through field trials with firefighting academies and active personnel will measure training transfer rates, learning curves, and task performance metrics comparing our SCBA-integrated HMD with standard VR setups.

Fourth, we envision expanding multimodal integration, combining the SCBA-HMD with haptic and sensory systems—thermal suits and force-feedback nozzles—for richer training environments.

Finally, wireless and standalone operation will be explored for untethered use in diverse training scenarios.

This research represents significant advancement in user-centered, high-fidelity VR training systems for emergency responders, laying groundwork for safer, more effective preparedness through realistic simulation technologies.

Acknowledgments

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